

F1 Wet Weather Strategy & Performance Analysis

Python | SQL | Excel | Power BI

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Executive Summary

This project investigates how wet-weather conditions relate to Formula 1 tyre strategy and driver race performance. Five wet-weather Grand Prix were analysed by combining race-level rainfall measures with driver-level tyre stint and finishing position data. Python was used for data preparation and exploratory analysis, SQLite for structured SQL analysis, and Power BI for interactive reporting.

Research Questions

1. How does rainfall severity relate to intermediate-tyre usage?
2. Does longer rainfall duration correspond with greater driver position change?
3. Which drivers achieved the largest gains in wet-weather races?
4. How does strategy vary between races with different rainfall conditions?

Data & Methodology

The analysis covers the 2019 German GP, 2021 Emilia-Romagna GP, 2023 Dutch GP, 2023 Monaco GP and 2024 British GP. Weather data was summarised using total rainfall, maximum hourly rainfall and number of hours with rain. Driver strategy data was transformed into measures including grid position, finish position, position change, tyre changes, first intermediate/wet lap, intermediate laps and intermediate-tyre share.

Stage	Purpose	Tools
Data Preparation	Clean, transform and combine race/weather data	Python / Pandas
Exploratory Analysis	Correlations, race summaries and visualisations	Python / Matplotlib
Database Analysis	Joins, rankings and driver/racequeries	SQLite/SQL
Reporting	Interactive dashboard and filtering	Power BI

Findings

Race	Rain	Max/hr	Rain hrs	Average Position Change	Average Intermediate Tyre Usage
2019 German GP	11.7	7.0	21	+5.83	0.667
2021 Emilia-Romagna GP	0.7	0.2	5	+1.33	0.394

2023 Dutch GP	3.4	0.7	12	+2.00	0.264
2023 Monaco GP	6.9	4.3	9	+0.83	0.302
2024 British GP	2.0	0.7	8	+0.83	0.228

1. Rainfall and tyre strategy

The strongest relationships in the exploratory correlation analysis were between rainfall measures and intermediate-tyre usage. Total rainfall and intermediate share had a correlation of approximately 0.675; rainfall hours and intermediate share were approximately 0.681; and maximum hourly rainfall and intermediate share were approximately 0.687. These results are directionally consistent: more persistent or intense wet conditions tended to coincide with a greater proportion of laps completed on intermediate tyres.

2. Rainfall and position change

The relationship between weather and position change was weaker. Rain hours had a correlation of approximately 0.375 with position change, while total rainfall was approximately 0.321 and maximum hourly rainfall approximately 0.301. This is expected because finishing position is affected by many additional factors, including qualifying position, pace, incidents, pit-stop timing, penalties and safety-car periods.

3. Largest driver gains

Driver	Race	Position gain
Sebastian Vettel	2019 German GP	+18
Daniil Kvyat	2019 German GP	+11
Lance Stroll	2019 German GP	+11
Pierre Gasly	2023 Dutch GP	+9
Lando Norris	2023 wet-race dataset	+4

Note: the position-gain ranking is based on the project's calculated Position_Change field. The table highlights the largest observed gains rather than claiming that weather alone caused those results.

4. Race-level comparison

The 2019 German GP stands out both for rainfall duration and strategic volatility. It recorded 21 hours with rain and an average driver position change of +5.83 in the project dataset. By comparison, the 2024 British GP recorded 8 rain hours and an average position change of +0.83. This illustrates how race outcomes can vary substantially even when both events involve wet conditions.

Analytics, Dashboard & Recommendations

1. SQL analysis

A SQLite database was created with two analytical tables: `racess`, containing race-level rainfall measures, and `race_strategy`, containing driver-level strategy and performance measures. SQL queries were used to rank position gains, compare rainfall across races, join weather with driver strategy, summarise race-level performance, identify gains during heavier rainfall and compare drivers across multiple wet races.

2. Power BI dashboard

The final dashboard brings the analysis together through KPI cards, scatter plots, a race-level position-gain chart, a driver strategy/performance table and Year, Race and Driver slicers. The dashboard is designed to let a user move from high-level race conditions to individual driver strategy and performance.

Dashboard component	Analytical purpose
Wet Races Analysed	Shows dataset coverage
Average Position Change	Summarises driver movement
Largest Position Gain	Highlights strongest individual result
Average Intermediate Usage	Tracks strategy response to wet conditions
Rainfall vs Intermediate Usage	Explores weather–strategy association
Rain Duration vs Position Change	Explores weather–performance association
Driver Strategy & Performance	Supports driver-level comparison

3. Limitations

The sample contains only five races, so the analysis should be treated as exploratory rather than a definitive statistical study. Race-level rainfall variables are repeated across drivers within each race, meaning driver-level correlation values should not be interpreted as five-driver-independent

observations. The analysis also does not fully control for safety cars, red flags, incidents, penalties, team pace or pit-lane conditions.

Future development

- Expand the dataset to a larger selection of wet-weather races from 2018 onwards.
- Introduce lap-by-lap weather and track-condition data.
- Add safety-car, red-flag, pit-stop and incident variables.
- Compare team-level strategic decisions and pit-stop timing.
- Develop predictive models for tyre-selection or strategy outcomes.
- Create a more advanced Power BI experience with drill-through analysis.

Conclusion

The project demonstrates an end-to-end analytics workflow from data preparation through SQL modelling and business-style visual reporting. The main finding is that rainfall intensity and duration show a substantially stronger association with intermediate-tyre usage than with position change. The results also demonstrate how extreme wet-weather races can create large strategic opportunities, exemplified by Vettel's 18-place gain at the 2019 German GP.

Portfolio: Python notebook • SQLite analysis • Power BI dashboard • GitHub repository